

Steps in whole genome sequencing/GWAS

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Background on WGS/GWAS

DNA

ACTGACCTAGATCAGTGTAGCGATCGTATACGAGACCGATTTCATCGGCAT



transcription

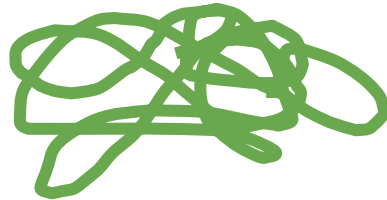
RNA

AUCAGUCGAUCACCGAU

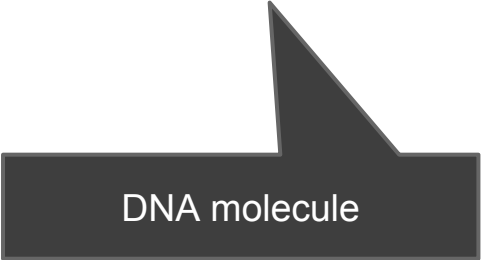


translation

protein



ATGGGAATTCACGAATTCCTAGACCTGCCCCGGAAACCTACCGCCGCG

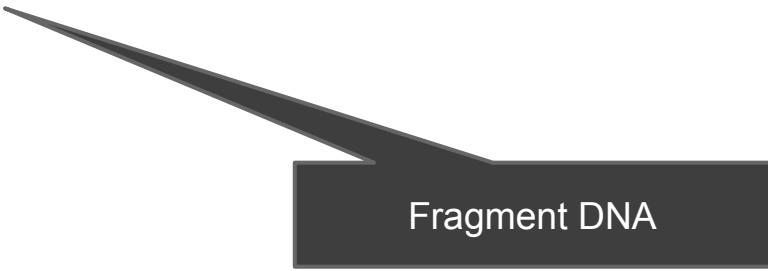


DNA molecule

ACCTGCCCCGGAAACCTACC

GCCGCG

ATGGGAATTCACGAATTCCTAG



Fragment DNA

The diagram illustrates a DNA insertion event. A dark grey arrow points from a rectangular box labeled 'Fragment DNA' to a specific location within the sequence 'ATGGGAATTCACGAATTCCTAG'. The arrow points to the space between the 'A' and 'C' of the 'ACGAATTC' subsequence, indicating where the fragment is being inserted.

ATGGGAATTCACGAATTCCTAG

Sequence DNA



ACACCTGCCCCGGGAAACC

ACACCTGCCCCGGGAAACC

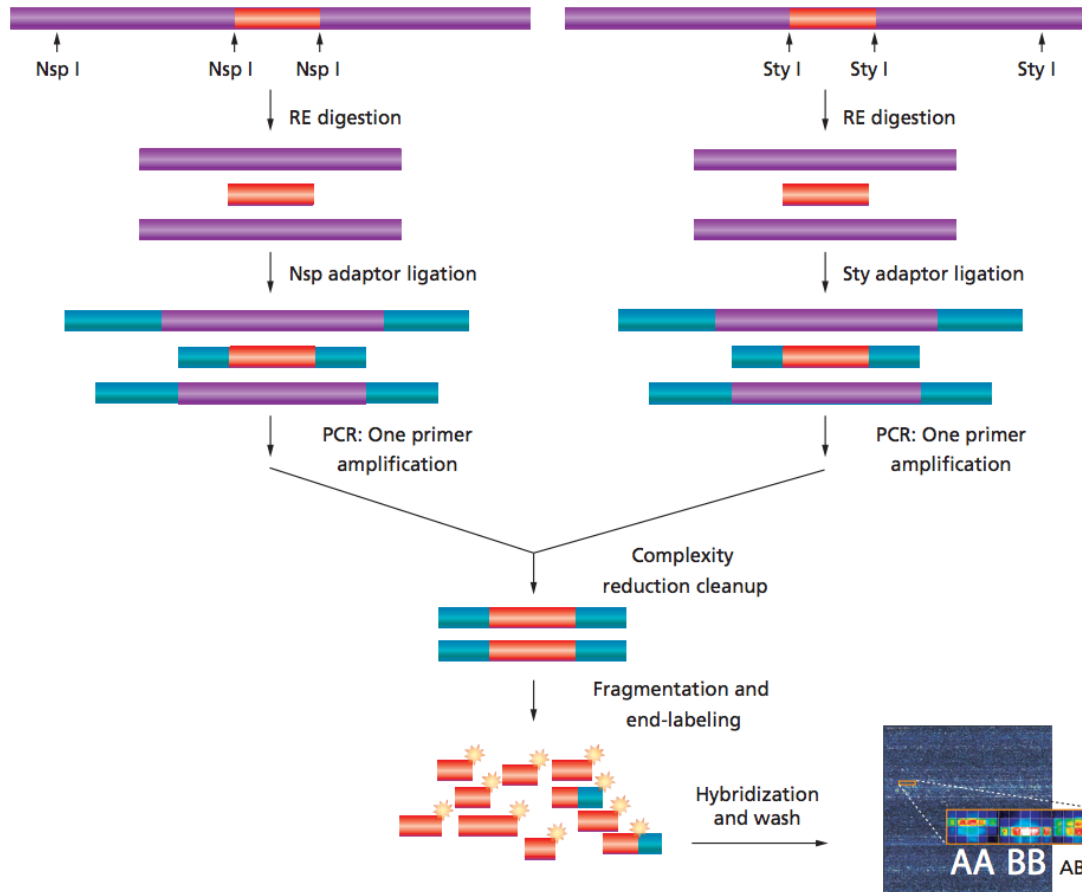
TCCTAGACCTGCCCCGG

AATTCCTAGACCTGCCCC

CGAATTCCTACACCTG

ATGGGAATTCACGAATTCCTAGACCTGCCCCGGGAAACCTACCGCCGCG

Genome



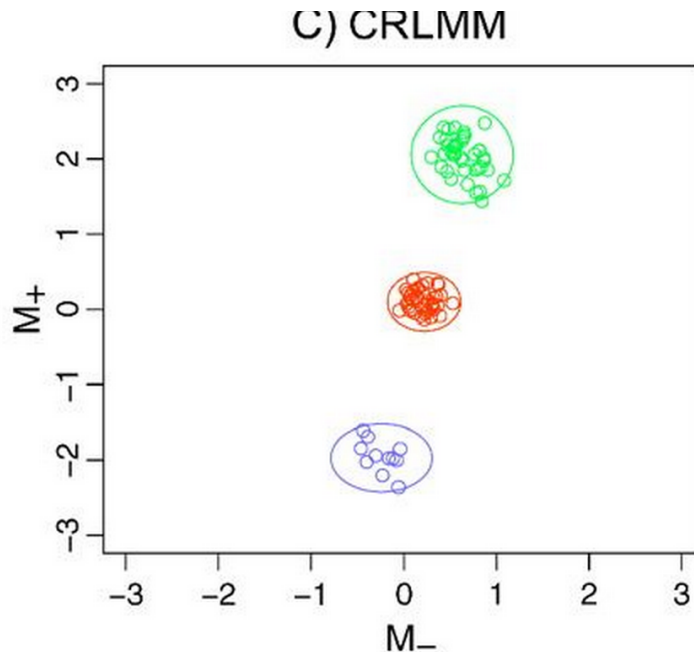
Steps

1. Variant identification
2. Population stratification correction
3. Statistical tests
4. Examining local region
5. Annotation

Step 1: Variant identification (SNP chip)

Software:

- [crlmm](#)



<http://www.ncbi.nlm.nih.gov/pubmed/17189563>

Step 1: Variant identification (sequencing)

Software:

- [freeBayes](#)
- [GATK](#)

		Variant Region				Variant Region			
Reads	Ref	TACCGAT	CATTGGATCA	CGATTCC...GCATTGC	AAAAAAA-	GACCGCA			
	TACCGAT	CATTGGATCA	CGATTCC...GCATTGC	-AAAAAA-	GACCGCA				
	ACCGAT	TATTGCATCG	CGATTCC...GCATTGC	-AAAAAA-	GACCGCA				
	ACCGAT	CATTGGATCA	CGATTCC...GCATTGC	AAAAAA-A	GACCGCA				
	ACCGAT	TATTGGATCG	CGATTCC...GCATTGC	-AAAAAAA	GACCGCA				
	CCGAT	C-TTGGATCA	CGATTCC...GCATTGC	AAAAAAA-	GACCGCA				
	CCGAT	CATGGGATCA	CGATTCC...GCATTGC	AAAAAAA A	GACCGCA				
		...	...	...	...	...			
Observed Haplotypes	CATTGGATCA		x8	(A) ₇		x10			
	TATTGGATCG		x9	(A) ₆		x7			
	CTTGGATCA		x1	(A) ₅		x1			
	CATGGGATCA		x1	(A) ₈		x1			
	...			...					

<https://github.com/ekg/freebayes>

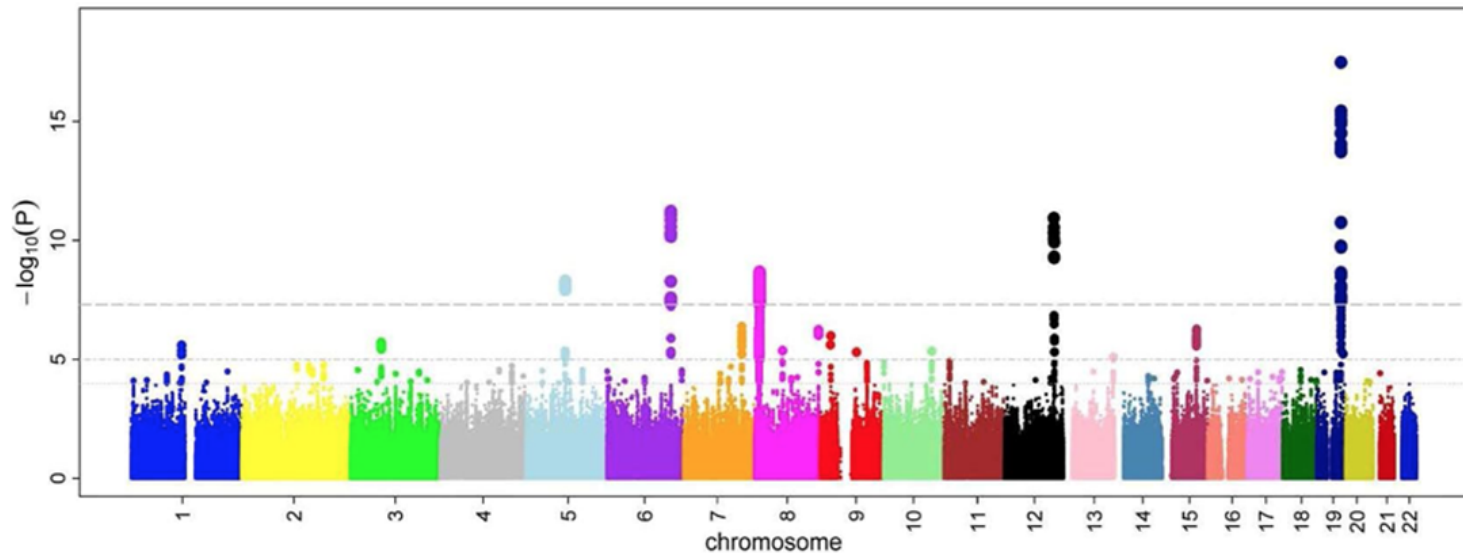
Software:

-
- a**
- PCA plot showing the first two principal components (PC1 and PC2) of genetic data. The plot displays numerous labeled points representing different populations, colored by country. An inset map of Europe in the top right corner shows the geographical distribution of these populations, with colors corresponding to the points in the PCA plot. The populations are distributed across Europe, with a high density in Central and Western Europe, and a few outliers in the North and South.

Step 3: Statistical tests

Software:

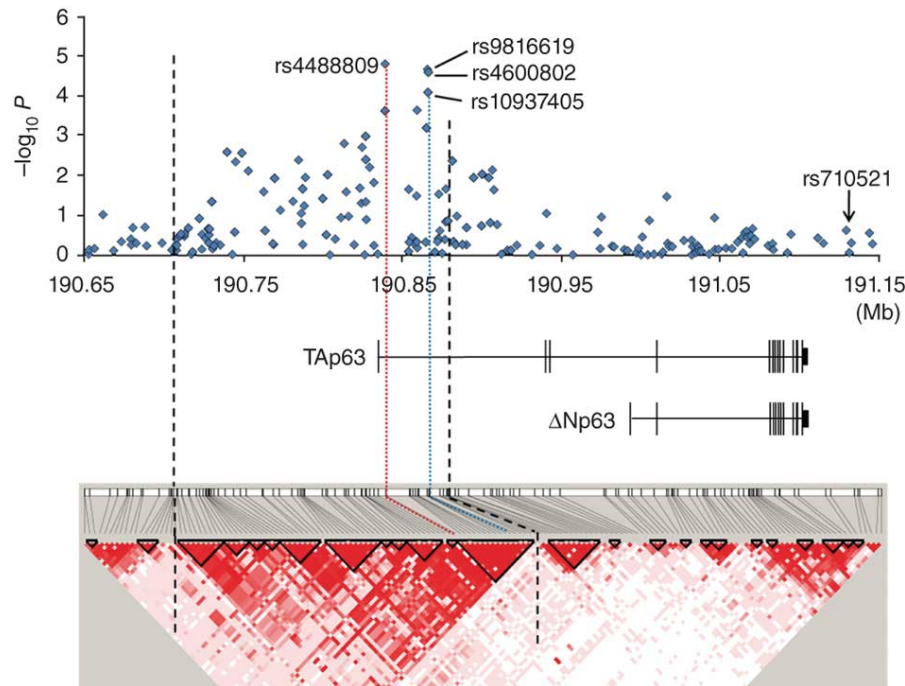
- [PLINK](#)
- [snpStats](#)



Step 4: Examine local region

Software:

- [PLINK](#)
- [Annotating Genomic Variants Workflow](#)



Step 5: Annotation

Software:

- [CADD](#)
- [variantAnnotation](#)
- [Annotating Genomic Variants Workflow](#)

